

NEURAL BELLMAN OPERATORS

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We study certified computation in a stopped consumption–portfolio economy with costly preference adjustment. In the executed financed policy class, direct L-BFGS-B remains more accurate and cheaper than the neural alternatives. We develop a synchronous policy-improvement operator with complete optimizer-state rollback, a finite-work quotient-polling theorem under explicit oracle and smoothness conditions, and second-order interval verification for current-state neural policies. Expanded, disjoint calibration brackets all six first-order configurations; a moving-transport control retains historical Jacobians and Adam moments. Eighteen new online decisions include five actual rejections. In a separate stopped economy with unspanned risk and an independently known value, unchanged neural policies obtain whole-state–time regret certificates below .001 on 32,768 cells, rather than 262,144. Exact rational polynomial verification provides a stronger classical reference. The original economy and its uniform .01 objective are retained: its present bound is 7.181835, and neither neural efficiency dominance nor a continuous stopped-gradient certificate is asserted.

KEYWORDS: Stochastic control, Bellman operators, Preference adjustment, Validated numerics, Policy improvement, Computational economics.

1. INTRODUCTION

Economic policy computation requires more than a small optimization loss. A proposed policy must be feasible in the economy being studied, its payoff must refer to the intended stopping rule, and its error must hold on the state set to which the conclusion applies. These requirements become distinct when a neural parameterization, numerical financing, and a continuation-value approximation are used together. Our objective is a Neural Bellman Operator for the stopped consumption–portfolio economy defined below, with a uniform regret tolerance of .01 on its original state–time domain. This objective is unchanged throughout the revision.

The present contribution is a set of constructive steps toward that objective, with the evidence attached to each step. First, we couple proposal generation to certified policy improvement. A rejected proposal restores not only a deployed policy but also the optimizer’s parameters, first and second moments, and iteration counter. This changes the next proposal. The distinction matters: certifying a completed trajectory and subsequently selecting its best checkpoint does not implement such an operator. Second, we give a derivative-free quotient-polling theorem with explicit accuracy and work requirements for the payoff oracle. It provides a route from verified improvement to a stationarity statement without treating an unverified proposal gradient as an economic gradient. Its smoothness and vanishing-width hypotheses are stated separately from the properties established for the current stopped-payoff checker.

Third, we strengthen whole-domain verification. In a stopped investment economy with an unspanned productivity shock, an exact Hamiltonian completion converts policy regret into an integral of squared action errors. Centered second-order interval bounds certify two fixed neural policies on the entire state–time box, including arbitrary restart times. They meet a .001 tolerance on 32,768 cells. The same policies previously required 262,144 cells with the inherited first-order verifier. A separate exact rational Bernstein calculation certifies the direct polynomial policy without the floating derivative-coefficient shortcut identified in our audit. These are improvements in verification of fixed policies, not claims of better training.

The classical comparison is essential to the economic interpretation. The expanded calibration study stored in the repository before this revision uses disjoint tuning and evaluation seeds and includes a moving historical-transport control. All six first-order selections lie strictly inside the expanded rate grid with deterioration at a larger rate. Nevertheless, direct L-BFGS-B has both a lower regret bound and strictly higher certified payoff than neural Adam in every matched central-state comparison. The manufactured stochastic benchmark likewise favors a classical approximant. These results rule out a neural-efficiency claim for the tested designs; they do not alter the policy class, economic model, or target that the paper seeks to solve.

The original current-state experiment has a complete-domain bound of 7.181834580823298, compared with the previously reviewed 7.241462443133396. Neither number establishes the .01 objective. Moreover, an improved verification bound is not, by itself, evidence of improved policy payoff. The stopped-gradient bridge also remains uninstantiated for the original proposal objective. We therefore maintain separate records for a proved identity, a certified numerical result, an optimization diagnostic, and a research objective. The new stochastic result is not substituted for the original economy.

Our framework draws on verification and approximation methods for continuous-time control (Kushner and Dupuis, 2001), the distinction between feasible policies and bounds (Brown, Smith, and Sun, 2010), and the numerical treatment of economic dynamic models (Judd, 1998, Brumm and Scheidegger, 2017). The proofs and complete numerical records from earlier revisions remain available without deletion. The companion historical annex reproduces all 93 pages of the reviewed R24 article and all 58 pages of its technical supplement; their source files are retained at their original paths. This article presents the current argument, while the new supplement supplies the complete calibration, checkpoint, verification, and provenance records.

2. THE STOPPED ECONOMY AND THE NUMERICAL OBJECTIVE

The horizon is $T = 1$. The state is $(u, x) \in D = (1.2, 2.8) \times (.5, 2)$, where u indexes preferences and x is wealth. Controls $a = (c, \theta, p)$ take values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Under the original Brownian filtration,

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = ((.02 + .06p_s)x_s - c_s) ds + .2p_s x_s dZ_s^x, \quad d\langle Z^u, Z^x \rangle_s = -.25 ds. \quad (2)$$

The adjustment price is $k \in [.5, 8]$; the central comparison uses $k = 2$. Let τ be the first exit from D , capped at 1. Running utility, terminal preference, and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1 - s). \quad (4)$$

For an admissible progressively measurable control,

$$J_k^a(t, u, x) = \mathbb{E}_{t, u, x} \left[\int_t^\tau e^{-.04(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-.04(\tau-t)} g(\tau, u_\tau, x_\tau) \right],$$

$$V_k = \sup_{a \in \mathcal{A}} J_k^a. \quad (5)$$

Admissibility includes a well-defined stopped state process. The bounded state and control ranges keep payoffs bounded; neural feedbacks used in the current-state calculations are smooth before stopping. We do not assume that the unknown value is classically differentiable.

For a smooth test function v , the controlled discounted residual is

$$\begin{aligned} \mathcal{R}_k^a v = & v_t + \theta v_u + ((.02 + .06p)x - c)v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \ell_k(u, c, \theta) - .04v. \end{aligned} \quad (6)$$

The numerical objective concerns a single current-state actor $a = \pi(t, u, x)$ and a uniform bound for all starting times and all states in the original domain. An initial-state financed expert library and an auxiliary economy are distinct objects, even when their numerical bounds are smaller.

PROPOSITION 1—Stopped residual certificate: *Suppose a smooth candidate v and an admissible policy π satisfy $|v - g| \leq b$ on the stopping boundary, $|\mathcal{R}_k^\pi v| \leq e$ throughout the state–time domain, and*

$$\sup_{a \in A} \mathcal{R}_k^a v - \mathcal{R}_k^\pi v \leq q.$$

Then, at every starting state,

$$0 \leq V_k - J_k^\pi \leq 2b + (2e + q) \frac{1 - e^{-.04(1-t)}}{.04}. \quad (7)$$

The same argument applies to separate upper and lower test functions with their respective one-sided residual and boundary inequalities, and to time-dependent error envelopes integrated up to 1.

The statement requires inequalities over a complete cover, not losses at training nodes. The inherited full-state certificate uses the stronger slabwise, sign-separated implementation, rather than replacing it by the constant bound in (7).

2.1. State information in the financed implementation

The central experiments use sixteen time slabs and a 1–16–16–3 tanh generator with 355 parameters. Its 48 node outputs determine a financed family with 47 economically distinct coordinates: a common intercept displacement is offset by the financing root. The decoder bounds consumption slopes and preference drift and reconstructs wealth through exact conditional pricing. This construction supplies admissible candidates; it is not a numerical identity between a general current-state policy and a financed library.

For expert i , write its conditional wealth as $X_i = F_i(s, Y_s)$, where

$$dY_s = .02Y_s ds - .3Y_s dZ_s^x, \quad p_i = -\frac{1.5Y_s F_{i,y}(s, Y_s)}{F_i(s, Y_s)}.$$

With fixed initialization weights $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, aggregate consumption and preference adjustment are weighted sums and portfolio weight is wealth-weighted. On the financed reachable set, $x = \sum_i \lambda_i X_i$ and the aggregate dynamics satisfy (2). The deployed implementation is Markov in (t, u, x, Y, λ) , not generally in (t, u, x) alone.

A causal auxiliary state with no additional randomness does not enlarge the class of progressively measurable controls under the original filtration. Conversely, an original admissible

control can ignore the auxiliary state. Thus unrestricted optimal values agree when the augmented admissible class is precisely this causal class. The equality says nothing about optimality of the restricted expert library. Off-manifold admissible extensions do not provide numerical accuracy at arbitrary memory states. The retained R24 regional bound below .004281 concerns $t = 0$, $u \in [1.98, 2.02]$, $x \in [1.24, 1.27]$, with initialized memory and nine experts. Its economic scope is kept separate from the whole-domain current-state objective.

3. CERTIFIED POLICY IMPROVEMENT AND NUMERICAL GEOMETRY

Let $y \in \mathbb{R}^{47}$ parameterize the financed quotient, and let $F(y)$ denote the exact stopped payoff at a specified initial state. This section distinguishes the proposal objective, the certified payoff oracle, and the state of the optimizer. No identity between their gradients is presumed.

3.1. Synchronous improvement with complete rollback

An optimizer state s contains economic coordinates, neural or chart-carrier parameters, first and second moments, and the iteration counter. Let $\Phi_h(s)$ be a block of proposal updates under a declared resource budget. The proposed policy receives a certified payoff enclosure $I(y) = [L(y), U(y)]$. If the checker fails, it returns no enclosure and the proposal is not accepted. The improvement operator is

$$\mathcal{T}(s) = \begin{cases} \Phi_h(s), & L(y(\Phi_h(s))) > U(y(s)), \\ s, & \text{otherwise.} \end{cases} \quad (8)$$

On rejection the entire saved state is restored. The block size and next learning rate are controlled by an external, recorded schedule; a rejected candidate's moment state is not used by the next block.

PROPOSITION 2—Verified improvement: *If every accepted enclosure contains the exact payoff of the corresponding admissible policy, every accepted update in (8) strictly increases that payoff, and the deployed sequence is nondecreasing. If acceptance instead requires $L(y') > U(y) + \sigma h^2$ with $\sigma > 0$, $h > 0$, and $F \leq M$, then there are at most $\lfloor (M - F(y_0))/(\sigma h^2) \rfloor$ accepted updates at that fixed scale.*

This is a safety and finite-improvement statement, not a convergence claim for Adam. If intervals overlap, failure to certify an improvement is not evidence that the candidate is worse. Restoring the optimizer is consequential only when a rejection occurs; the earlier R25 execution accepted all 24 blocks. Section 5.2 therefore executes a declared stress schedule that exercises the rejection branch, retains full states, and separately continues rejected states as diagnostic shadow runs.

3.2. Precision-adaptive payoff decisions

The absence of a stopped-gradient certificate does not prevent exact-payoff comparisons when sufficiently narrow payoff enclosures are available. Let the width of each policy enclosure at level j be at most $w_j = w_0 2^{-j}$. For a prescribed improvement threshold $s \geq 0$, compare

$$[L(y') - U(y), U(y') - L(y)]$$

with s . Accept if its lower endpoint exceeds s , reject the threshold claim if its upper endpoint is below s , and otherwise refine or return *undecided* at the resource cap.

LEMMA 1—Finite separation of a nonzero payoff margin: *Assume the enclosures above are valid at every level. If $\gamma = |F(y') - F(y) - s| > 0$, the threshold decision is certified whenever $2w_j < \gamma$. It suffices to use at most*

$$\max\{0, \lceil \log_2(2w_0/\gamma) \rceil + 1\}$$

refinements. At zero margin no finite separation is guaranteed.

The current original-economy checker supplies valid intervals at its executed settings. We have not proved that all its components shrink to zero along every relevant policy trajectory. That additional oracle property is a hypothesis of Lemma 1, not an empirical conclusion from two quadrature rules. In particular, rigorous secants are not pointwise derivative enclosures.

3.3. A quotient-polling theorem with explicit verification work

A finite-work result can be obtained without differentiating the proposal quadrature. Consider a direct poll in the economically identified coordinates. Let $\mathcal{D}$ be a finite set of unit directions with positive cosine measure

$$\mu = \inf_{g \neq 0} \max_{d \in \mathcal{D}} \frac{g'd}{\|g\|} > 0.$$

For the coordinate poll $\mathcal{D} = \{\pm e_i : 1 \leq i \leq 47\}$, $\mu = 47^{-1/2}$. At scale h , accept a feasible trial $y + hd$ only if its certified lower payoff exceeds the incumbent upper payoff by σh^2 . A complete unsuccessful poll tests every direction.

ASSUMPTION 1—Poll regularity and oracle accuracy: *On all poll segments under consideration, F is continuously differentiable with L -Lipschitz gradient and is bounded above by M . Every prescribed poll point and segment is feasible and inside this regularity region; accepted iterates remain in it. The payoff oracle returns a valid interval of width at most κh^2 at each point, at cost at most $W(\kappa h^2)$. Here $\sigma > 0$ and $\kappa \geq 0$ are fixed.*

THEOREM 1—Accuracy–work bound for a certified quotient poll: *Under Assumption 1, a complete unsuccessful poll satisfies*

$$\|\nabla F(y)\| \leq \frac{\sigma + 2\kappa + L/2}{\mu} h. \quad (9)$$

Starting at y_0 and continuing fixed-scale polls until the first unsuccessful one requires no more than

$$(|\mathcal{D}| + 1) \left(1 + \left\lceil \frac{M - F(y_0)}{\sigma h^2} \right\rceil \right) \quad (10)$$

payoff-oracle evaluations, with their total verification work bounded by (10) times $W(\kappa h^2)$. Taking $h = \mu\epsilon/(\sigma + 2\kappa + L/2)$ yields an ϵ -stationarity bound with $O(\epsilon^{-2})$ polls, apart from the explicitly retained oracle cost.

The estimate is a conditional numerical-method theorem, not an assertion that the original stopped objective already meets Assumption 1. It does not establish a global optimum, a rate for Adam, or a dimensional advantage of neural coordinates. A missing feasible direction at a boundary is not counted as an unsuccessful complete poll. In this revision the theorem is

checked on an exact quadratic example; a 47-dimensional certified poll was not run on the original economy.

The poll itself depends on quotient policies and certified payoffs, not on an equivalent parameter chart of the neural generator. This is an exact form of chart independence for its acceptance decisions. Chart independence of this component must not be confused with invariance of Adam or L-BFGS-B in neural weights. A complete implementation may use neural proposals first and the quotient poll as a safeguard, but its evaluation cost must include every verification required by (10).

3.4. Historical neural transport and an executed control

For a raw output map $f : \mathbb{R}^{355} \rightarrow \mathbb{R}^{48}$, let $J_n = Df(\vartheta_n)$ and $g_n = \nabla_y L(f(\vartheta_n))$. With bias-corrected Adam first moment,

$$\hat{m}_n = \sum_{r=1}^n a_{nr} J'_r g_r, \quad a_{nr} = \frac{(1 - \beta_1) \beta_1^{n-r}}{1 - \beta_1^n},$$

and $D_n = \text{diag}(\sqrt{\hat{v}_n} + \epsilon)$, Taylor's theorem gives

$$f(\vartheta_{n+1}) - f(\vartheta_n) = -\eta_n \sum_{r=1}^n a_{nr} J_n D_n^{-1} J'_r g_r + \epsilon_n, \quad (11)$$

$$\|\epsilon_n\| \leq \frac{H}{2} \eta_n^2 \|D_n^{-1} \hat{m}_n\|^2, \quad (12)$$

provided Df is H -Lipschitz on the update segment. Current $J_n D_n^{-1} J'_n$ alone omits the historical transport. The retained static tangent, diagonal, and whitening charts do not reproduce (11).

The additional direct arm uses quotient coordinates $z \in \mathbb{R}^{47}$ and a separate 355-parameter chart carrier ϑ . At each step it computes $B_n = Q' J_n$, pulls the direct payoff gradient back through B'_n , updates parameter-space Adam moments, and pushes the resulting parameter displacement forward through B_n . Its economic displacement is

$$\delta z_n = B_n \delta \vartheta_n - \eta_n 10^{-6} \nabla_z L, \quad \|\delta z_n\| \leq .5, \quad (13)$$

where a common cap factor rescales the economic and carrier displacements. The carrier is updated in parameter space, while z is updated linearly. The regularizer and cap were fixed before evaluation. The method retains historical pullbacks, adaptive second moments, and the current pushforward, but intentionally omits the finite-step nonlinear output remainder in (11). It is therefore a moving-transport control, not an exact neural-trajectory replica. Its 355 carrier coordinates and two 355-dimensional moment arrays are real costs; calling it a memory-free 47-variable method would be incorrect.

3.5. What financing-gradient control requires

Let $P(y, a) = B_0$ be the exact financing equation, P_N its numerical approximation, and $a(y), a_N(y)$ their roots on a common bracket. Let $H(y, a)$ and $H_N(y, a)$ denote exact and numerical payoff functions before root substitution. Suppose $P_a, P_{N,a} \geq d_* > 0$ and $|P - P_N| \leq e_P$, so $|a - a_N| \leq \delta = e_P/d_*$. At the respective roots, let

$$\|H_{N,y} - H_y\| \leq E_A, \quad |H_{N,a} - H_a| \leq E_B, \quad \|P_{N,y} - P_y\| \leq E_C, \quad |P_{N,a} - P_a| \leq E_D.$$

Each E includes fixed-offset approximation error and the applicable offset-Lipschitz allowance times δ . The derivative factor bounds are explicitly two-sided in the numerical/exact distinction:

$$\max\{|H_a|, |H_{N,a}|\} \leq M_B, \quad \max\{\|P_y\|, \|P_{N,y}\|\} \leq M_C. \quad (14)$$

PROPOSITION 3—Reduced-gradient perturbation: *Under these assumptions and differentiability of both implicit roots,*

$$\|\nabla_y H_N(y, a_N(y)) - \nabla_y H(y, a(y))\| \leq E_A + \frac{E_B M_C + M_B E_C}{d_*} + \frac{M_B M_C E_D}{d_*^2}. \quad (15)$$

The bounded proposal extension clips preference before running-utility evaluation. Its integrability and derivative implementation are not an identification of its gradient with that of (5). Neither this revision nor the imported R25 experiment supplies verified values for every term in (15) for the original stopped payoff along the executed optimizer paths. We instead report certified payoff decisions and finite secants as such. The theorem removes an ambiguity in the previous derivative-factor statement; it does not manufacture the missing constants.

4. WHOLE-DOMAIN VERIFICATION IN AN UNSPANNED-RISK STOPPED ECONOMY

To test stochastic propagation, stopping, and current-state restart without the original financed decoder, consider a separate investment economy on

$$(t, x, z) \in [0, 1] \times [.5, 2] \times [-1, 1], \quad a \in [0, 2.2], \quad p \in [0, 1],$$

with independent Brownian motions W, B :

$$dX = (.02X + a + .04pX) dt + .25pX dW, \quad dZ = .3 dB. \quad (16)$$

The B shock is not spanned by the risky wealth exposure. Let stopping occur at first exit from the interior box or time 1. Set $A(t) = .8 + .2t$ and supply the smooth witness

$$v(t, x, z) = A(t)(\log x + .1xz - .1z^2). \quad (17)$$

At every stopping face, settlement equals v . Running reward is $b(t, x, z) - a^2/2$, where the model's transfer is

$$b = .04v - v_t - .02xv_x - \frac{.3^2}{2}v_{zz} - \frac{v_x^2}{2} + \frac{.04^2v_x^2}{2(.25)^2v_{xx}}. \quad (18)$$

All these coefficients are bounded on the closed box, and $v_{xx} = -A/x^2 < 0$. This is a manufactured model: the transfer is chosen to make the value independently known. The experiment validates policy approximation and continuous verification in a stochastic stopped setting; it is not an exercise in discovering an unknown value function or calibrating an empirical economy.

THEOREM 2—Exact stochastic value and restart regret: *For the model (16)–(18), the optimal current-state actions are*

$$a^*(t, x, z) = A(t)(1/x + .1z), \quad p^*(t, x, z) = .64(1 + .1xz). \quad (19)$$

They lie in the declared action bounds, and the unrestricted stopped value is v . For every admissible current-state policy $\pi = (a, p)$,

$$v(t, x, z) - J^\pi(t, x, z) = \mathbb{E}_{t, x, z} \int_t^\tau e^{-.04(s-t)} \frac{(a - a^*)^2 + .0625A(s)(p - p^*)^2}{2} ds. \quad (20)$$

Consequently, a uniform upper bound $\bar{d}$ on the integrand gives

$$0 \leq v - J^\pi \leq \bar{d} \frac{1 - e^{-.04(1-t)}}{.04} \quad (21)$$

at every original state of this auxiliary economy and at every restart time, without stored initialization weights or a financing root.

The state–time statement is stronger than a list of simulated starting states. It follows from a complete cover and a stopped verification identity. The tabulated start times below are evaluations of the same uniform formula; no interpolation between those times is used in the proof.

4.1. Centered second-order action certificates

Let $e_j = \pi_j - \pi_j^*$ be an action error on a rectangular cell with center m and half-widths r_i . Interval propagation supplies point enclosures of $e_j(m)$ and $\partial_i e_j(m)$, and whole-cell enclosures of $\partial_{ik} e_j$. Denote valid upper absolute bounds by E_j , G_{ij} , and H_{ikj} .

THEOREM 3—Second-order cell enclosure: *If e_j is twice continuously differentiable on the cell, then throughout it*

$$|e_j| \leq E_j + \sum_i G_{ij} r_i + \frac{1}{2} \sum_i H_{iij} r_i^2 + \sum_{i < k} H_{ikj} r_i r_k. \quad (22)$$

The minimum of this bound and a valid first-order mean-value enclosure remains valid. Substituting such componentwise error bounds into the nonnegative quadratic integrand in (20), maximizing over a complete cover, and applying (21) yields a whole-state–time certificate.

For each tanh or sigmoid layer, the implementation propagates

$$D(\phi \circ q) = \phi'(q)Dq, \quad D^2(\phi \circ q) = \phi'(q)D^2q + \phi''(q)Dq Dq'.$$

All affine operations and transcendental enclosures use the inherited directed interval arithmetic. Saved binary64 network weights are interpreted as exact real coefficients; displayed economic decoder constants are exact rationals. The certificate therefore concerns the real-valued policy represented by those coefficients, not an unspecified hardware rounding path at deployment. The control scales are exact 2.2 and 1. A separate floating-point comparison with automatic differentiation tests the implementation of the chain rule; it does not replace the interval proof.

Both neural policies are the unchanged R25 3–12–12–2 policies. They were fitted using the known actions on a $5 \times 9 \times 7$ tensor rule. R26 does not retrain them. It verifies nested covers of sizes 512, 4096, and 32768, retaining the unsuccessful coarse certificates. Every endpoint and center of these covers is dyadic. The code saves every cell, its action-error bounds, its quadratic deficit bound, and a file digest. No sampled cell is substituted for an unverified remainder of the domain.

4.2. Exact rational verification of the classical reference

The retained direct policy approximates $1/x$ by a degree-eight polynomial $P(\xi)$, $\xi = (x - 1.25)/.75$, uses $a = A(P(\xi) + .1z)$ with projection onto $[0, 2.2]$, and uses the exact portfolio expression in (19). Projection cannot increase distance to the feasible optimum. The polynomial coefficients are the unchanged stored binary64 coefficients, interpreted exactly as rational numbers.

The inherited direct verifier formed derivative coefficients with the ordinary floating multiplication $i c_i$ before treating them as point intervals. That operation does not itself enclose the exact product of the stored coefficient and i . We do not retroactively label the old derivative-coefficient computation as a directed enclosure. Instead we remove the derivative calculation from the current certificate.

LEMMA 2—Exact residual-polynomial certificate: *Let $R(\xi) = ((5 + 3\xi)/4)P(\xi) - 1$. On a cell $[a, b] \subset [-1, 1]$, express $R(a + (b - a)s) = \sum_{i=0}^n \beta_i \binom{n}{i} s^i (1 - s)^{n-i}$ with exact rational coefficients. Then*

$$\sup_{\xi \in [a, b]} |P(\xi) - 1/x| \leq \frac{\max_i |\beta_i|}{(5 + 3a)/4}. \quad (23)$$

With E the largest bound over a complete cover, the direct policy's regret is at most $\frac{1}{2}E^2(1 - e^{-.04(1-t)})/.04$ for every starting state and time.

The computation uses 64 one-dimensional cells. The t, z dependence is handled analytically through $A \leq 1$, and portfolio error cancels exactly. The resulting bound is $1.2978908373252045 \times 10^{-7}$. Its small value is not a claim about superiority of a general classical Bellman solver; it is the correct strong baseline for this structured, known-value problem. In particular, it prevents a verifier improvement from being reported as neural-policy superiority.

5. NUMERICAL EVIDENCE AND ECONOMIC SCOPE

The evidence has two distinct chronologies. We import the completed R25 experiment at commit `c78d934` without changing its scientific files. Its protocols precede its execution. The present R26 protocol was recorded locally after examining those results and before the new stress tests and fixed-policy verification. It is not described as a remotely preregistered or independently held-out study. The new seeds 26101, 26102 are diagnostic stress seeds; the fixed stochastic policies retain their original seeds 25201, 25202.

5.1. Expanded calibration and held-out comparison

The imported experiment expands the learning-rate grid to

$$\{.005, .02, .08, .32, 1.28, 5.12, 20.48\}.$$

At seeds 25001, 25002, every arm receives 200 objective/gradient calls per rate. Selection uses the mean certified endpoint payoff lower bound; infeasible endpoints are ineligible, and the fixed tie rule selects the smaller rate. The selected rates are .08 for neural and whitening Adam and .32 for direct, diagonal, static tangent, and moving-transport Adam. All six selections have a worse larger-rate trial and are strictly below the grid ceiling. This repairs the earlier endpoint-censoring

TABLE I
ORIGINAL ECONOMY, CENTRAL STATE: IMPORTED HELD-OUT COMPARISON

Method	Regret upper	Calls	Generation seconds	Checking seconds
Neural Adam	0.0019264	400.0	1.144	13.836
Direct Adam	0.0019367	400.0	1.016	13.887
Diagonal Adam	0.0019329	400.0	1.039	13.961
Static tangent Adam	0.0020835	400.0	1.023	13.923
Whitening Adam	0.0179096	400.0	1.041	13.958
Moving transport Adam	0.0019275	400.0	1.687	13.897
Neural L-BFGS-B	0.0096083	243.0	0.671	13.840
Direct L-BFGS-B	0.0019235	123.0	0.307	13.891

The state is $(t, u, x) = (0, 2, 1.25)$, $k = 2$, proposal orders $(8, 16)$. Regret is the worst certified deployed bound over two evaluation seeds; work columns are seed means. These are financed policies, not whole-domain current-state actors. Checking sums scheduled checkpoint verification; it is not a measured online latency. Carrier construction and common preprocessing, and any common dual construction, are not included in the generation column. The full run records remain available.

problem for this finite design, not a proof that all schedules or all hyperparameters have been optimized.

Evaluation uses seeds 25101, 25102 and proposal orders $(8, 16)$ and $(12, 24)$, without retuning. Each first-order arm has a 400-call cap. L-BFGS-B has the same cap, with its declared line search and tolerances; early termination is retained rather than padded into fictitious calls. The 84 tuning trajectories, 32 held-out trajectories, six earlier online trajectories, and separate reference studies have different inferential purposes. Their sum is bookkeeping, not a replication count for one effect.

Direct L-BFGS-B remains the strongest tested central-state solver. This is not merely a difference in common upper bounds: each matched interval for neural Adam payoff minus direct L-BFGS-B payoff is strictly negative. At the coarse rule the two intervals are approximately $[-2.3223, -1.5593] \times 10^{-6}$ and $[-3.2620, -2.4991] \times 10^{-6}$. The fine-rule comparisons have the same sign. These are certified differences, not significance tests across a population of seeds.

Moving transport substantially closes the gap between the static tangent arm and neural Adam, but does not beat the direct L-BFGS-B frontier. Because its carrier incurs differentiation and storage, its generation time is charged rather than presented as costless quotient preconditioning. The retained R24 equivalent-chart and neural L-BFGS-B failures are preserved. Neither a static-chart failure nor this moving-control result establishes an irreducible neural advantage.

5.2. Actual online rejection and recovery

The new stress schedule has three blocks: 200 calls at the selected base rate, 50 calls at 20.48, and 100 recovery calls. The recovery rate is halved after a rejected stress block. This deliberately aggressive block tests the mechanism, not the solver frontier. Neural, direct, and moving-transport arms use the original central economy and the unchanged stopped checker. Every proposal, incumbent, decision, and optimizer snapshot is retained.

Across six trajectories, 13 proposals are accepted and five rejected. Four rejected aggressive candidates produce separate shadow continuations from the rejected optimizer state. All four shadow final states differ from the actual rolled-back trajectories. The moving arm accepts both aggressive proposals, but one later recovery proposal is rejected. Thus rejection is not asserted for a branch never entered. Exact equality of the canonical before/after optimizer-state

TABLE II
R26 SYNCHRONOUS STRESS EXECUTION IN THE ORIGINAL CENTRAL ECONOMY

Seed	Arm	A/R	Regret upper	Gen.	Check	Coupled
26101	Neural	2/1	0.0019257	1.057	8.176	9.269
26101	Direct	2/1	0.0019471	0.941	8.056	9.012
26101	Moving transport	2/1	0.0019339	1.646	8.001	9.674
26102	Neural	2/1	0.0019266	1.058	7.896	8.989
26102	Direct	2/1	0.0019469	0.943	7.994	8.952
26102	Moving transport	3/0	0.0019310	1.543	7.862	9.433

A/R denotes accepted/rejected blocks. Times are seconds from the new local serial execution. Check includes the initial incumbent certificate. Coupled time is the measured interleaved run from initial evaluation through the final decision; it includes bookkeeping but excludes initial model construction, inherited common dual construction, and the separately timed shadow continuation. All arms use 350 block gradient calls; these stress runs are not the held-out comparison in Table I.

TABLE III
FIXED-POLICY STOCHASTIC CERTIFICATES ON THE SAME CELL COVERS

Seed	Cells	First order	New certificate	Check seconds
25201	512	0.15838620	0.12137299	0.145
25201	4,096	0.00939394	0.00423980	1.123
25201	32,768	0.00137310	0.00083434	8.848
25202	512	0.20476383	0.17534400	0.144
25202	4,096	0.01128653	0.00515876	1.109
25202	32,768	0.00133294	0.00078841	8.846

Both methods certify the same stored neural policy over $[0, 1] \times [.5, 2] \times [-1, 1]$ in the manufactured, unspanned-risk stopped economy. The new certificate takes the minimum of two valid componentwise error bounds. Times include both same-cover calculations and cell-file writing in the local execution. They are not a hardware-normalized speedup relative to the earlier CI run.

serializations is checked at every rejection, including moments and step counters. No shadow certificate is used to replace the production incumbent.

These executions establish synchronous certification and actual state restoration. They do not establish an Adam convergence rate or show that the stress schedule improves production efficiency. The monotonicity theorem applies to the executed acceptance decisions; the finite-work polling theorem concerns the separately defined poll and its stated hypotheses.

5.3. Whole-state-time reference verification

Table III compares first-order and second-order error enclosures on identical cells for fixed neural policies in the auxiliary stochastic economy. Each attempted cover is retained, including those above the target.

The target .001 is attained at 32,768 cells for both policies, whereas the retained first-order stopping rule required 262,144. The reduction is eightfold in cell count at the same declared tolerance, not an eightfold measured runtime gain. The final bounds, .000834337 and .000788415, are valid at all starting times; the earlier finer certificates may of course be numerically tighter. The policies have not changed. The exact rational direct bound, approximately 1.298×10^{-7} , remains much stronger. A successful stochastic neural certificate therefore demonstrates nontrivial verified approximation and restart coverage, not a neural accuracy/work victory.

5.4. *Original full-state progress and stopped-payoff sensitivity*

The imported continuation experiment retains the original economy, current-state actor, full domain, and .01 target. It uses 85 gradient calls across four blocks and delivers an all-start-time bound of 7.181834580823298. Relative to 7.241462443133396, the bound decreases by about .059628, or .823%. The remaining bound is approximately 718 times the target. No separately certified actor-payoff improvement is recorded. Neither a local financed bound nor a manufactured-model certificate is used to change this conclusion.

The imported secant experiment certifies five original stopped-payoff policies along a raw-coordinate interpolation between neural Adam and direct L-BFGS-B outputs. Of four adjacent secant intervals, only the first is strictly positive, approximately $[5.332, 8.383] \times 10^{-6}$. The remaining three contain zero. These comparisons rigorously concern finite policy differences; they do not bound every continuous derivative, prove an optimizer ordering robust to all surrogate errors, or instantiate Proposition 3. They provide an actionable comparison primitive while the full stopped-gradient bridge remains a separate mathematical and computational task.

5.5. *Reproducibility and preservation*

The new source contains a protocol, all three attempted covers for each fixed neural policy, exact rational Bernstein coefficients on all 64 cells, every online state snapshot, shadow records, machine-readable publication tables, and checks for the completed Hamiltonian identity, derivative propagation, exact polynomial conversion, complete cover, acceptance/rollback invariants, and historical preservation. The single root revision index is updated, with its old bytes archived. No historical scientific source or result is overwritten. The new response identifies the disposition of all twelve blocking findings, all six technical findings, and all eight requested next steps.

6. CONCLUSION

A Neural Bellman Operator for the original stopped preference-adjustment economy remains the organizing objective. This revision adds an executed certification-coupled proposal mechanism, a conditional finite-work quotient-polling theorem, a moving historical-transport comparison, and sharper whole-domain verification in a stopped unspanned-risk economy. The experiments distinguish a real optimizer rollback from checkpoint replay and a verifier gain from a policy gain. They retain a strong direct benchmark and the adverse neural outcomes rather than selecting away the relevant comparison.

The next successful original-economy calculation must supply a current-state actor with a complete-domain certificate near the declared .01 tolerance and a separate payoff-improvement statement when such a statement is made. A proof that the original checker supplies arbitrarily narrow policy intervals, or instantiated stopped-gradient constants, would connect the verified-improvement component more tightly to the optimization analysis. The present results do not claim those conclusions. They provide reproducible methods, proofs, and evidence with which to pursue them, while preserving the original economic problem and every prior derivation for further scrutiny.

APPENDIX A: PROOFS

A.1. *Stopped verification*

PROOF OF PROPOSITION 1: Apply stopped Itô's formula to $e^{-.04(s-t)}v(s, u_s, x_s)$, localizing first if needed. On the bounded stopped domain the stochastic integral has zero expectation

under the stated admissibility and smoothness conditions. Thus

$$J_k^a - v(t, u, x) = \mathbb{E} \int_t^\tau e^{-.04(s-t)} \mathcal{R}_k^a v(s, u_s, x_s) ds + \mathbb{E}[e^{-.04(\tau-t)}(g - v)(\tau, u_\tau, x_\tau)].$$

For every a , $\mathcal{R}_k^a v \leq e + q$, giving $J_k^a \leq v + b + (e + q)C_t$, where $C_t = (1 - e^{-.04(1-t)})/.04$. For π , $\mathcal{R}_k^\pi v \geq -e$, giving $J_k^\pi \geq v - b - eC_t$. Take the supremum over a and subtract. One-sided test functions follow from the same identity; deterministic time envelopes may be integrated to 1 because $\tau \leq 1$. Q.E.D.

A.2. Improvement, oracle precision, and polling

PROOF OF PROPOSITION 2: Validity of the two enclosures gives $F(y') \geq L(y') > U(y) \geq F(y)$ on acceptance. A rejection restores the original policy and optimizer state. With forcing term σh^2 , each accepted payoff gain is strictly larger than that quantity. The sum of these gains is bounded by $M - F(y_0)$, yielding the stated conservative integer bound. No assertion about gradient convergence is needed. Q.E.D.

PROOF OF LEMMA 1: For a valid enclosure of width at most w_j , its lower endpoint is no less than the true value minus w_j , and its upper endpoint is no greater than that value plus w_j . Consequently the lower and upper difference endpoints differ from the true payoff difference by at most $2w_j$ in the relevant direction. If $2w_j < \gamma$, the whole difference interval lies strictly on the correct side of the threshold. The displayed refinement count ensures this strict inequality. If the margin is zero, no positive finite width need separate it. Q.E.D.

PROOF OF THEOREM 1: At an unsuccessful poll, $L(y + hd) \leq U(y) + \sigma h^2$ for every $d \in \mathcal{D}$. Interval widths bounded by κh^2 imply

$$F(y + hd) - F(y) \leq (\sigma + 2\kappa)h^2.$$

The Lipschitz-gradient inequality on the feasible segment gives

$$F(y + hd) - F(y) \geq h \nabla F(y)' d - \frac{L}{2} h^2.$$

Combining these inequalities, maximizing over d , and using the cosine measure yields (9). Each successful poll increases payoff by more than σh^2 . The payoff upper bound therefore allows at most the number of successes in Proposition 2 before an unsuccessful complete poll. At most $|\mathcal{D}| + 1$ oracle evaluations per poll, including a conservative incumbent reevaluation, gives (10). Multiplying by the uniform oracle cost gives the work bound. The chosen h gives the stationarity tolerance. All feasibility, regularity, and oracle requirements are precisely those in Assumption 1; failure of any requirement does not imply stationarity. Q.E.D.

A.3. Transport and financing perturbations

For completeness, the first-moment recursion $m_n = \beta_1 m_{n-1} + (1 - \beta_1) J'_n g_n$ with zero initialization expands to the coefficients a_{nr} after bias correction. Taylor expansion of f along $-\eta_n D_n^{-1} \hat{m}_n$ yields (11); integrating the Lipschitz Jacobian difference along the segment bounds the remainder by $H \|\delta \vartheta_n\|^2 / 2$. The executed carrier uses the corresponding sequence of pull-backs and pushforwards but advances economic coordinates linearly, as stated in (13).

PROOF OF PROPOSITION 3: The mean-value inequality and the lower bound on P_a imply $|a - a_N| \leq e_P/d_*$. Implicit differentiation gives $a_y = -P_y/P_a$ and $a_{N,y} = -P_{N,y}/P_{N,a}$. Hence the two reduced gradients are $H_y - H_a P_y/P_a$ and $H_{N,y} - H_{N,a} P_{N,y}/P_{N,a}$. Add and subtract products one factor at a time. The payoff-derivative and price-derivative differences contribute $(E_B M_C + M_B E_C)/d_*$; the reciprocal difference contributes at most $M_B M_C E_D/d_*^2$, since both denominators are at least d_* . The direct derivative difference contributes E_A . The root-shift allowances are already included in the E terms, so no offset error is counted twice. *Q.E.D.*

A.4. The stochastic reference and its interval certificates

PROOF OF THEOREM 2: The controlled residual of v in the reference economy is

$$v_t + (.02x + a + .04px)v_x + \frac{.25^2 p^2 x^2}{2} v_{xx} + \frac{.3^2}{2} v_{zz} + b - a^2/2 - .04v.$$

Complete the square in a and p , using $v_x = A(1/x + .1z)$ and $v_{xx} = -A/x^2$. With (18), the result is exactly

$$-\frac{1}{2}\{(a - a^*)^2 + .0625A(p - p^*)^2\}.$$

On the closed box, $.32 \leq a^* \leq 2.1$ and $.512 \leq p^* \leq .768$, so the unconstrained maximizers are admissible. The optimal controls are smooth on a neighborhood of the stopped domain. Stopped Itô's formula applies, with settlement equal to v on every stopping face. The residual is nonpositive for every admissible policy and zero for the displayed optimizer. Thus $V = v$ and the same formula with an arbitrary policy gives (20). Bounding the nonnegative integrand and using $\tau \leq 1$ proves (21) for every starting point and time. *Q.E.D.*

PROOF OF THEOREM 3: Write a point as $m + h$, with $|h_i| \leq r_i$. The second-order integral Taylor formula gives

$$e_j(m + h) = e_j(m) + D e_j(m) h + \int_0^1 (1-s) h' D^2 e_j(m + sh) h ds.$$

The integral weight is $1/2$. Absolute values and the symmetric Hessian expansion yield the diagonal half-weights and the single mixed terms in (22). Whole-cell interval Hessians cover every point on the segment. Both first- and second-order bounds are individually valid, so their componentwise minimum is valid. The Hamiltonian deficit is nondecreasing in the nonnegative squared-error bounds and in the positive coefficient A ; a complete cover and (21) then give the uniform result. *Q.E.D.*

PROOF OF LEMMA 2: The Bernstein basis is nonnegative on $[0, 1]$ and sums to one. Hence the polynomial lies between the smallest and largest exact coefficients β_i . The denominator $x = (5 + 3\xi)/4$ is positive and bounded below by $(5 + 3a)/4$ on the cell, proving (23). The exact action error before projection is $A(P - 1/x)$; projection onto the feasible interval cannot increase its distance from a^* . As $A \leq 1$ and the portfolio action is exact, the deficit is bounded by $E^2/2$. Apply Theorem 2. The conversion itself uses the identity that a local power coefficient q_k contributes $q_k \binom{i}{k} / \binom{n}{k}$ to Bernstein coefficient i , for $k \leq i$; every operation in this conversion is rational. *Q.E.D.*

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